

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI-590018, KARNATAKA



**A Project Report
on**

***Li-Fi Based Automatic Traffic Signal Control For
Ambulance***

*Submitted in partial fulfillment of the requirements for the VIII Semester of degree
of Bachelor of Engineering in Information Science and Engineering of
Visvesvaraya Technological University, Belagavi*

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An Institute with a difference

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CERTIFICATE

Certified that the project work entitled *Li-Fi Based Automatic Traffic Signal Control For Ambulance* has been successfully completed by **Amardeep Kumar (1RN19IS020)**, **Amartya (1RN19IS021)**, **Anushka Jaiswal (1RN19IS035)** and **Devendra Upadhyay(1RN19IS056)**, bonafide students of **RNS Institute of Technology, Bengaluru** in partial fulfillment of the requirements for the award of degree in **Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University, Belagavi** during the academic year **2022-2023**. The project report has been approved as it satisfies the academic requirements in respect of project work for the said degree.

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DECLARATION

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ABSTRACT

India's rapid development has led to ineffective traffic management, particularly in metropolitan cities such as Bengaluru, which affects emergency services such as ambulance transport. Li-Fi, a bi-directional visible light communication technology, can provide a feasible and efficient solution to traffic management issues. Li-Fi's advantages include fast and secure communication within a large area, particularly in traffic-dense areas. Li-Fi-based communication systems enable emergency vehicles, such as ambulances, firefighting trucks, and police vehicles, to communicate with each other and traffic signals, improving traffic flow and reducing response times.

This project explores the potential use of Li-Fi-based traffic management systems in improving ambulance services in India, particularly in metropolitan cities with heavy traffic congestion. Li-Fi technology's advantages, such as fast and secure communication and broader connectivity within traffic-dense areas, make it an ideal solution for enhancing traffic management. The proposed Li-Fi-based communication system enables emergency vehicles to communicate with each other and traffic signals, improving traffic flow and reducing response times. This project argues that the use of Li-Fi in emergency vehicle communication can significantly improve traffic management, potentially saving lives and improving emergency services' efficiency.

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Chapter 1

INTRODUCTION

The growth of metropolitan cities in India has resulted in severe traffic congestion, making it challenging for emergency services such as ambulances to reach their destination on time. The need for effective traffic management systems is crucial to ensure that emergency vehicles can navigate through congested roads and reach their destination promptly. One solution to this problem is the implementation of Li-Fi based automatic traffic signal control for ambulances.

1.1 Overview

Li-Fi is a wireless communication technology that utilizes visible light to transmit data, making it a promising alternative to traditional wireless communication methods that rely on radio waves. Li-Fi is an innovative wireless communication technology that uses visible light for data transmission, presenting a viable alternative to conventional wireless methods that rely on radio waves. It operates by modulating the intensity of LED light bulbs at high frequencies to transmit data. By using the visible light spectrum, Li-Fi can achieve high data transmission rates of up to several Gbps, which is much faster than traditional wireless technologies.

One potential application of Li-Fi technology is in traffic management systems for emergency vehicles, such as ambulances. By installing Li-Fi-enabled traffic lights, emergency vehicles can communicate with the traffic management system in real-time, enabling them to navigate through congested roads more efficiently. Li-Fi can also enable communication between emergency vehicles, which can help them coordinate their movements and avoid traffic congestion.

The implementation of Li-Fi-based automatic traffic signal control for emergency vehicles can have a significant impact on emergency services in India. With its ability to transmit data quickly and reliably in crowded urban areas, Li-Fi can provide a fast and secure method of communication for emergency services. This technology can help emergency vehicles reach their destination on time, thereby saving lives.

In addition to improving emergency services, Li-Fi-based traffic management systems can also reduce traffic congestion, improve traffic flow, and minimize delays. This can have a positive impact on the overall transportation system, making it more efficient and reliable.

Overall, Li-Fi technology has the potential to revolutionize the way we communicate and manage traffic in urban areas, providing faster and more reliable communication channels for emergency services and improving the efficiency of the transportation system as a whole.

Challenges Faced by Traffic System:

1. The traffic system faces various challenges that can affect its efficiency and effectiveness. Traffic congestion is one of the significant challenges that emergency services like ambulances face. The increasing population and urbanization have led to congested roads in many cities worldwide, making it challenging for emergency vehicles like ambulances to reach their destinations quickly. Traffic congestion can also cause delays in public transportation, resulting in decreased productivity and increased pollution.
2. Lack of proper infrastructure is another challenge faced by the traffic system. The traffic system's infrastructure is essential to ensure the smooth flow of traffic and reduce congestion. However, the lack of proper infrastructure, such as roads, highways, bridges, and tunnels, can lead to traffic congestion and delays.
3. Limited funding is another significant challenge faced by the traffic system. The traffic system requires significant investment in infrastructure, technology, and human resources. However, many governments and local authorities face budget constraints that limit their ability to invest in improving the traffic system.
4. Poor traffic management is another challenge faced by the traffic system. Poor traffic management, such as lack of real-time data, inefficient traffic signal timing, and inadequate planning, can lead to traffic congestion and delays. Effective traffic management is crucial to reduce congestion and ensure the smooth flow of traffic.
5. Safety concerns are also a significant concern for the traffic system. Traffic accidents can cause severe injuries, fatalities, and property damage. Proper safety measures, such as traffic signals, signs, and road markings, are essential to ensure the safety of all road users.

Potential of Li-Fi Technology for Traffic Management Systems:

Li-Fi technology provides a promising solution for traffic management systems for emergency vehicles. The technology uses visible light to transmit data, making it an attractive alternative to traditional wireless communication methods that rely on radio waves. Since Li-Fi operates in the visible light spectrum, it can achieve transmission rates of up to several Gbps, much faster than traditional wireless technologies, making it ideal for emergency response systems. By installing Li-Fi-enabled traffic lights, emergency vehicles can communicate with the traffic management system in real-time, providing them with the necessary information to navigate through congested roads more efficiently. The traffic management system can monitor the traffic flow and adjust the traffic lights accordingly to prioritize the emergency vehicle's route, ensuring a smooth passage through the congested roads. Moreover, Li-Fi can facilitate communication between emergency vehicles, allowing them to coordinate their movements

and avoid traffic congestion. This coordination can prevent multiple emergency vehicles from reaching the same location simultaneously, reducing the risk of accidents and ensuring that emergency services are available for all those in need. Li-Fi technology can also enable communication between emergency vehicles and hospitals, allowing hospitals to prepare for the patient's arrival by having the necessary equipment and staff ready. This communication can also provide real-time updates on the patient's condition, enabling hospitals to prepare the appropriate treatment plan before the patient arrives, potentially saving precious time in critical medical emergencies.

Feasibility of Li-Fi-based Automatic Traffic Signal Control:

Li-Fi technology offers several advantages for traffic management systems for emergency vehicles. However, several technical requirements and costs need to be considered before implementing the technology. One of the technical requirements of Li-Fi is the need for line-of-sight communication. Since Li-Fi uses visible light to transmit data, it requires an unobstructed line-of-sight between the transmitter and the receiver. Therefore, to implement Li-Fi-enabled traffic lights, it may require the installation of additional infrastructure such as light poles and positioning the traffic lights at a particular height for optimal communication. Another technical requirement is the need for high-intensity LED light bulbs. Since Li-Fi uses the intensity of LED light bulbs to transmit data, it requires high-intensity LED light bulbs to achieve the necessary data transmission rates. Therefore, retrofitting existing traffic lights with high-intensity LED light bulbs may be required, which can add to the cost of implementing the technology. Cost is another consideration when implementing Li-Fi-based automatic traffic signal control. The cost of retrofitting existing traffic lights with high-intensity LED light bulbs and installing additional infrastructure can be significant. However, the potential benefits of Li-Fi technology, such as faster emergency response times and improved traffic management, can outweigh the initial costs.

Li-Fi technology provides a promising solution for traffic management systems for emergency vehicles. By enabling real-time communication between emergency vehicles and the traffic management system, Li-Fi can help reduce traffic congestion and ensure the timely arrival of emergency services, potentially saving lives. However, several technical requirements and costs need to be considered before implementing the technology. The potential benefits of Li-Fi technology for traffic management systems make it an attractive option for cities facing traffic congestion and emergency response challenges.

Chapter 2

LITERATURE SURVEY

A literature survey or a literature review in a project report shows the various analyses and research made in the field of interest and the results already published, taking into account the various parameters of the project and the extent of the project. Literature survey is mainly carried out in order to analyze the background of the current project which helps to find out flaws in the existing system & guides on which unsolved problems we can work out. So, the following topics not only illustrate the background of the project but also uncover the problems and flaws which motivated to propose solutions and work on this project.

A literature survey is a text of a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literature reviews use secondary sources, and do not report new or original experimental work. Most often associated with academic-oriented literature, such as a thesis, dissertation or a peer-reviewed journal article, a literature review usually precedes the methodology and results sectional though this is not always the case. Literature reviews are also common in a research proposal or prospectus (the document that is approved before a student formally begins a dissertation or thesis). Its main goals are to situate the current study within the body of literature and to provide context for the particular reader. Literature reviews are a basis for researching nearly every academic field. A literature survey includes the following:

- Existing theories about the topic which are accepted universally.
- Books written on the topic, both generic and specific.
- Research done in the field usually in the order of oldest to latest.
- Challenges being faced and on-going work, if available.

Literature survey describes about the existing work on the given project. It deals with the problem associated with the existing system and also gives user a clear knowledge on how to deal with the existing problems and how to provide solution to the existing problems.

Objectives of Literature Survey

- Learning the definitions of the concepts.
- Access to latest approaches, methods and theories.
- It improves the quality of the literature survey to exclude sidetracks– Remember to explicate what is excluded.

Before building our application, the following system is taken into consideration:

[1] Modeling the Impact of VANET-Enabled Traffic Lights Control on the Response Time of Emergency Vehicles in Realistic Large-Scale Urban Area, 2013 (Hamed Noori)

The contribution of this paper is twofold: First, the effect of the changing traffic lights status to green for emergency cars is investigated by using traffic simulator (SUMO). Second, this paper uses OMNET++ (Network Simulator) in order to simulate the mentioned scenario as a VANET (with 802.11p standard) by using Veins framework to run SUMO and OMNET++ in parallel. This study has developed the Veins framework by adding a new module to OMNET++ to consider the traffic lights which are simulated in SUMO. Moreover this study has developed a new program written in Python which is connected to SUMO and controls the traffic simulation. This program uses SUMO to simulate a microscopic traffic (by considering every single vehicle movements) and also a city with intelligent traffic lights. Additionally, several statistics about traffic simulation is created for each car such as traveling time, waiting time, emissions, fuel consumption; or complete amount of car emissions in the street during the simulation, fuel consumption, number of vehicles and so on, for each street. Additionally, several statistics about traffic simulation is created for each car such as traveling time, waiting time, emissions, fuel consumption; or complete amount of car emissions in the street during the simulation, fuel consumption, number of vehicles and so on, for each street. This paper uses Manhattan realistic map to describe the mentioned program, then uses realistic map and realistic traffic demand of Cologne, Germany, to obtain a realistic and reliable result. [1]

[2] A GPS Based Traffic Light Pre-emption Control System for Emergency Vehicles, 2013 (Abubakr S. Eltayeb; Halla O. Almubarak; Tahani Abdalla Attia)

In this paper, the design and implementation of an automatic preemption traffic control system, based on the global positioning system satellites. It ensures the arrival of emergency vehicles to their destinations on the minimum time possible. The results indicate that the proposed system has an optimum solution for the delay time experienced on the emergency vehicles along their path. In the simulation program ISIS professional and an Atmega16 microcontroller was programmed to act and response as GSM modem proposed system is giving a unique and optimum solution to the main problem and other problems attached to the different pre-emption systems by suggesting the shortest path with the least obstacles taken from an updated electronic map to the driver of the emergency vehicle, which will result in reducing the emergency vehicles' travelling time to the desired destination and clearing the path in advance from other vehicles and pedestrians. Field experiment tests on the hardware

model on Othman Digna Street across Al-Jami'a Street at the city of Khartoum were successfully accomplished with fair results proved to be compatible with theoretic results to a great extend. [2]

[3] Implementing Intelligent Traffic Control System for Congestion Control, Ambulance Clearance and Stolen Vehicle Detection, 2015 (Rajeshwari S., Santhoshs Hebbar, Varaprasad Golla)

This paper presents an intelligent traffic control system to pass emergency vehicles smoothly. Each individual vehicle is equipped with special RFID tag(placed at a strategic location), which makes it impossible to remove or destroy. We use RFID reader, NSK EDK-125–TTL and PIC16F877A system-onchip to read the RFID tags attached to the vehicle. It counts number of vehicles that passes on a particular path during a specified duration. It also determines the network congestion, and hence the green light duration for that path. If the RFID-tag-read belongs to the stolen vehicle, then a message is sent using GSM SIM300 to the police control room. Also, when an ambulance is approaching the junction, it will communicate to the traffic controller in the junction to turn on the green light. This module uses ZigBee modules on CC2500 and PIC16F877A system-onchip for wireless communications between the ambulance and traffic controller. This module uses ZigBeemodules on CC2500 and PIC16F877A system-onchip for wireless communications between the ambulance and traffic controller. The prototype was tested under different combinations of inputs in our wireless communication laboratory and experimental results were found asexpected. [3]

[4] GPS and ZigBee Based Traffic Signal Preemption , 2016 (Vineeth Kodire, Sreebha Bhaskaran, Vishwas H.N.)

In this paper, whenever the emergency vehicle is approaching preemption messages are provided to the traffic signal unit. So these signals interrupt the normal operation of the traffic light and make it green for the emergency vehicle. It is a type of Vehicle-to-Infrastructure (V2I) communication. The proposed system uses GPS method to get the location data and employs ZigBee for communicating efficiently with the intersection module. There is no requirement of Line-ofSight employed in infrared based preemption methods. Also eliminate the false trigger occurring in sound (siren) based method due to noise and fading. Using this approach, it is possible to serve multiple emergency vehicles reaching the traffic intersection according to their distance from the intersection and accurately taking the decision to turn the signal green for the particular vehicle. There is no involvement of driver of the EV in anyway, it is fully autonomous and it can be used as a part of ITS.[4]

[5] Advanced Automation Control in an Ambulance under Emergency Condition , 2017 (Leila Sai Krishna; Samineni Vijay Chowdary; M. Pushpavalli; P. Sivagami)

In this paper, they have implemented a scheme which can control the traffic signals automatically in its path way and reduce the amount of traffic congestion at the signal. The ambulance is implemented with embedded system units, which finds the accident spot and delivers the spot to the close by ambulance through GPS .The ambulance guides the traffic lights in the roadway to the hospital and it also checks the next consecutive signal to shorten the time loss. The vehicle unit is equipped with vibration sensor to determine the vibration if it exceeds the level .The embedded unit sense the accident happens and its sends the neighborhood ambulance unit through wireless transmission. RFID reader is needed to extricate the ambulance from other vehicles.[5]

[6] Emergent vehicle tracking system using IR sensor, 2017 (Manjunath Managuli; Abhaya Deshpande; Sudha H Ayatti)

In this paper, they proposed a signal track division that use universal position method to find the exact position of an emergency van (ex. Ambulance, Fire exchange) and remote users get the information transmitted using global position system which also provides traffic signal track monitoring system i.e., during object motion IR sensors detect the ambulance, pass the message to main controller which reflects the LED bulb to glow leading to closing of all the signals track, and immediately the emergent vehicle pass the signal track. The main principle of this scheme is to plan an emergent tracking method. The proposed system is implemented based on real time embedded system for this implementation they used ATMEG328P controller which is the heart of the system. The IR sensor is used to sense the data which is processed by the controller and sent the data to emergent tracking system through GPS.[6]

[7] Intelligent Ambulance with Automatic Traffic Control, 2020 (P. Devi; S. Anila)

This paper have come up with the solution of “Intelligent automatic traffic control for ambulance ”. The proposed system creates a android app that connects both the ambulance and the traffic signal station using cloud network. This system makes uses RFID (radio frequency identification) technology to implement the Intelligent traffic signal control. The basic idea behind the proposed system is, if the Ambulance halts on the way due to a traffic signal, RFID installed at the traffic signal tracks the RFID tagged ambulance and sends the data to the cloud. After the acknowledgment for the user through the mobile app, the particular signal is made Green for some time and after the ambulance passes by, it regains its original flow of sequence of signaling If, this scheme is fully automated, it finds the ambulance spot,

controls the traffic lights. This system control the traffic lights and save the time in emergency periods.[7]

[8] Leveraging Emergency Response System Using the Internet of Things, 2020 (Luz A. Magre Colorado; Jorge Franco Ibañez; Juan Carlos Martinez-Santos)

This paper propose a hardware/software system that allows the emergency vehicles to “talk” to the traffic-lights nearby to prioritize them over the other cars while maintaining the roads safe. This system should take the current infrastructure (surveillance camera network and the traffic-light system) to coordinate the best route. This system should connect the surveillance camera network and the traffic-light system to coordinate the best route. Our main objective is to develop a hardware/software system that allows the emergency vehicles to “talk” with the traffic-lights around to prioritize them over the other cars while keeping the safety on the roads. This system should connect both the surveillance camera network and the trafficligh system to coordinate the best route. This system should connect both the surveillance camera network and the trafficligh system to coordinate the best route. The proposal is an Emergency Vehicle Signal Preemption System in cooperatives Vehicle Infrastructure System. However, the approach will complement the systems of intelligent traffic lights based on readings of transit through the checkpoints (cameras infrastructure system). Using the technology of the QR code, mobile networks, and GPS for the control and automation of the traffic lights in favor of the emergency vehicles that need to transit through high congestion road zones using the infrastructure explained. After all, it was going too fast for the system to catch it. We use the GSM network to communicate any updates, GPRS to monitor the location of the vehicle, and checkpoints to confirm how the execution of the operation is going.[8]

[9] Li-Fi technology in traffic light, 2017 (V.K.G. Kalaiselvi; A. Sangavi; Dhivya)

The proposed paper aims in using the Wi-Fi technology and enabling communication of vehicles with the traffic light system in order to prioritize the vehicles and change the signals accordingly rather than by a process of pre-defined order or by manual order. Traffic lights already use LED lighting, so that this proposed system may seem easy to implement. Traffic lights already use LED lighting, so that this proposed system may seem easy to implement. Sending data through siren lights in an ambulance and fire extinguishers to a traffic light control system and switching the signal in order to allow faster and non-interrupted transport. Here is already a kind of system in play for GPS navigational systems, but the traffic lightscan be used in updating drivers using basic information or streaming video directly from news broadcasts i.e, about the accidents or delays up ahead.[9]

[10] Efficiency of opportunistic cellular/LiFi traffic offloading, 2017 (Vishnu Raj RS, Athi Narayanan S, Kamal Bijlani)

In this paper, our main aim is to discuss the traffic offloading efficiency of opportunistic cellular/LiFi HetNet. In addition, the effect of LiFi-signal blocking is taken into consideration. Furthermore, the effects of user existence in LiFi cell and dynamic variation of user densities on LiFi offloading efficiency are considered. The rest of this paper is organized as follows. In Section 2, state- and timing-diagram for different offloading cases are explained and mathematical expressions for the distributions of different timing parameters are given. In Section 3, the LiFi offloading efficiency is investigated numerically. We have investigated traffic offloading efficiency of cellular/LiFi HetNet. A framework of opportunistic cellular/LiFi offloading and timing diagrams for different disjoint cases have been illustrated. Statistical distributions for different times have been explored. Our results reveal that LiFi blocking reduces the traffic offloading efficiency. In addition, the users' density has a positive effect on LiFi offloading efficiency in opportunistic cellular/LiFi HetNet.[10]

[11] Li-Fi Based Smart Traffic Network, 2018 (A. Correa; A. Hamid; E. Sparks)

In this paper, The Smart Traffic System will utilize Li-Fi to avoid using the already crowded radio air waves. Our system will also eliminate a problem faced by the Tesla Autopilot System, a currently popular autonomous car technology. This system is currently designed to be used on limited-access highways, and not in urban streets as it is unable to improvise in construction areas and locations with no lane markings. Our system would require a light transmitter be placed on each traffic sign to send a signal to cars that alert the system of the position of each sign so the car can determine the action required by each sign. As traffic signals already emit light, they would not require an additional transmitter. They would only need to be reprogrammed in order to change the flash rate of the existing lights. This project was simulated using Intel's Altera Cyclone V FPGA. [11]

[12] Li-Fi Based Automatic Traffic Signal Control for Emergency Vehicles, 2018 (R Shanmughasundaram; S Prasanna Vadanam; Vivek Dharmarajan)

In this paper, In this paper, a V2V communication system is proposed in which the headlight and tail light of a vehicle are made as Li-Fi transmitter and Li-Fi receiver respectively. Using the proposed system, when a traffic signal receives an alert regarding an emergency vehicle, the traffic signal would immediately turn green. Thus, decreasing the waiting time of emergency vehicles in traffic dense lanes. In this paper, a single hop scenario was implemented. The time taken for the alert message to propagate from an emergency vehicle to

a non-emergency vehicle and then to the traffic signal control is approximately one second. This time can be further reduced by implementing the same functionality using optimized code. The time can also be reduced by designing better modulation and demodulation circuits which supports higher data rates. Alert message can also be encrypted to make it secure. [12]

[13] Highway Navigation Using Light Fidelity Technology, 2015 (Abhishek Patni, Bhavini Mishra, Harsh Aditya, Yogesh Kumar and Rohit Sharma)

In this paper, they presented light is everywhere street lamp, traffic light and meeting hall. Imagine that all these light bulbs are high speed wireless transmitter that connects either human to human or systems with systems. Light fidelity or we can say LIFI, a new WIFI is based on the very recent intelligence of light emitting diode (LED) technology as the source of lightning. LIFI is a fully networked, bidirectional high speed wireless connection system. Unlike wireless networks. LIFI networks do not rely on any fixed frequency spectrum which is very limited. Instead, LIFI is based on visible light spectrum which provides the spectrum which is more than 10,000 times to radio wave to deliver desired data. These unique characteristics of LIFI pose a number of challenges for the implementation of capacity, efficiency, availability and security in the wireless network system. These unique characteristics of LIFI pose a number of challenges for the implementation of capacity, efficiency, availability and security in the wireless network system. In this Topic, we study the LIFI use in highway navigation system thus understanding the vulnerabilities and how LIFI will use full in future. Further, we discuss the various issues in WIFI and explore approaches to solve the problems in communication. [13]

[14] Intelligent traffic signal control system for ambulance using RFID and cloud, 2017 (J. Saradha, G. Vijayshri and T. Subha)

The main concept behind the paper is to provide a smooth flow for the ambulance to reach the hospitals in time and thereby minimizing the delay caused by traffic congestion. The Microcontroller based RFID system is used to alter the traffic lights upon its arrival at traffic light junction which would save a lives at critical time. Radio Frequency Identification is a tiny electronic gadgets that comprise of a small chip and an antenna. The small chip is embedded with information's about patient's status and the ambulance current lane. The RFID reader located at the traffic signal reads these information from the RFID locator installed at the ambulance. To avoid unnecessary traffic signal changes, we cross refer the ambulance current location and patient's condition using mobile app registered by the ambulance driver.. In case of network failure, the RFID takes the whole control. [14]

[15] Li-Fi Communications in Smart Cities for Truly Connected Vehicles, 2021 (Jamshaid Iqbal Janjua; Tahir Abbas Khan; Md Saeed Khan; Mehwish Nadeem)

This paper speaks about the future communication of Li-Fi and its implementation in connected vehicles in smart cities. The term Li-Fi is used for high-speed VLC which offers a higher bandwidth, connectivity and secure communication between vehicle-to-vehicle communication. Besides the strengths of Li-Fi technology, the aspect of weaknesses of Li-Fi technology are also discussed and summarized. Vehicles and traffic infrastructure communications can gain benefits that are wide-ranging, by implementing Li-Fi technology as a communication medium. We have proposed a Li-Fi “Transmitter” and “Receiver” model that enable safe inter-operative network wireless communications between automotive and implementation architecture is presented for Li-Fi communication between automotive and traffic infrastructure for vehicle integration. This paper also demonstrates a VLC based communication system in which data is sent from Li-Fi transmitter to a Li-Fi receiver using visible light communication (VLC) as a medium. [15]

Summary

Various communication protocols like GSM, RFID, IR, ZigBee etc. have been considered to implement emergency vehicle based traffic control. GSM based communication a system has been extensively used for cellular communication. The infrastructure needed for GSM communication is high compared to other communication protocols. Even though basic infrastructure for GSM is available, the use of this technology for V2V communication requires high bandwidth. RFID based V2V communication do not require line of sight. RFID based traffic control can detect emergency vehicles if it is in the frequency range of the traffic control. The prime disadvantage of RFID based V2V communication is low security offered where RFID messages are prone to unauthorized access. The metallic chassis of the vehicle can also cause problems to the RFID transceiver. IR based emergency vehicle detection is Unreliable as IR signals require line of sight and sunlight can cause interference with IR signals. ZigBee based traffic management is a good alternative when compared to other communication protocols mentioned above. It offers wide range of communication does not require line of sight and is comparatively economical. However, to implement V2V communication or traffic management using ZigBee involves restrictions on the number of nodes per controller. Various research papers have proposed the use of Li-Fi for emergency vehicle based traffic control systems. It does not need any infrastructure like GSM. It offers a much secure communication compared to RFID. It is relatively economical compared to ZigBee. It does not use the congested 2.4 GHz frequency like ZigBee.

Chapter 3

ANALYSIS

3.1 Existing System and Limitations

The existing traffic signal control systems can be broadly classified into three categories: manual control, fixed-time control, and actuated control. Manual control involves a human operator who changes signal timing based on judgment of traffic flow, but is susceptible to human error leading to inefficiencies and delays. Fixed-time control uses pre-determined and fixed signal timing regardless of traffic volume, which is easy to implement but can cause delays and congestion during peak traffic periods. Actuated control is a method that utilizes sensors to detect the presence of vehicles or pedestrians at the intersection and adjust the signal timing based on real-time traffic flow. However, this method is not specifically focused on emergency vehicles and their need for priority in traffic signal control.

Therefore, alternative methods such as GPS-based pre-emption control systems or Li-Fi-based automatic traffic signal control systems have been proposed to provide priority to emergency vehicles. GPS-based systems use GPS signals to detect the approach of emergency vehicles and adjust the signal timing accordingly. However, these systems have limitations in urban areas where GPS signals can be blocked or weakened. On the other hand, Li-Fi-based systems use visible light communication to transmit data between the ambulance and the traffic signal controller, enabling faster and more accurate detection of the approaching emergency vehicle. The Li-Fi-based automatic traffic signal control system can provide real-time priority to emergency vehicles, reducing response time and potentially saving lives. However, this system is still in its early stages of development, and its implementation may require significant infrastructure and cost.

3.2 Problem Identification

Cities with continually growing populations and the same infrastructure have been facing the problem of increasing traffic congestion. As a result, it has become increasingly challenging for emergency vehicles such as ambulances to reach their destinations within the recommended response time of 8-10 minutes, which is critical in cases of medical emergencies like cardiac arrest or stroke. This problem can lead to loss of lives, and it is essential to address it with a feasible solution. The successful implementation of Li-Fi-based automatic traffic signal control for ambulances could significantly improve emergency response times, leading to better patient outcomes.



Figure 3.1 Ambulance stuck in traffic jam.

There are specific critical points in every city where traffic slows down, making it difficult for emergency vehicles to navigate through. This can result in a delayed response time, which can have severe consequences, especially in cases of medical emergencies. In highly populated metropolitan cities like Bengaluru, traffic jams can be a major issue, which can affect the ambulance services. Therefore, there is a need to identify an efficient solution that can provide timely and secure connectivity for emergency vehicles.

The existing traffic signal control systems are not designed to prioritize emergency vehicles like ambulances, firefighting, and police vehicles. The current systems are based on fixed timers or sensors and do not take into account the priority of emergency vehicles. This lack of prioritization results in delayed response times, which can be critical in saving lives during medical emergencies. Therefore, there is a need for a new system that can provide a more efficient and effective solution for traffic signal control, especially for emergency vehicles.

3.3 Objectives

The objective of the proposed project is to implement a Li-Fi based automatic traffic signal control system for ambulances that reduces response time of emergency vehicles, especially ambulances, by providing them with priority passage through traffic signals. The project aims to improve overall traffic flow in the city while providing emergency vehicles with priority passage, minimizing risks of accidents and loss of lives due to delayed response times of emergency vehicles.

The features that we are implementing in this project are as follows:

- Real-time modification of traffic signal direction in response to incoming ambulances.
- Integration with hospitals to provide up-to-date information on available beds to the ambulance crew.

3.4 Proposed System

Li-Fi, or Light Fidelity, is a wireless communication technology that uses light to transmit data. It works by using LED lamps that can be switched on and off at very high speeds, which is not noticeable to the human eye. These light signals can be received by a photo-detector, which converts them back into electrical signals.

In the proposed system, LED-based Li-Fi transmitters will be installed in emergency vehicles, which will emit signals containing information about the vehicle's location and direction. The Li-Fi receivers installed in the traffic signal poles will then receive these signals and convert them into electrical signals.

The traffic signal controller will analyze these electrical signals and use the information to adjust the signal timing to prioritize the passage of the emergency vehicle. For instance, if the Li-Fi receiver detects an ambulance approaching from the east, it will adjust the traffic light timings to give priority to the ambulance and clear the path for it.

The proposed system involves connecting hospitals to a central server in order to facilitate communication between hospitals and ambulances. Hospitals with available beds will send signals indicating their availability to the ambulance. The ambulance, in turn, will receive this signal through its Arduino Uno microcontroller, which will be connected to a Wi-Fi module.

This system has the potential to significantly improve the efficiency of the healthcare system, particularly during times of high demand. By connecting hospitals and ambulances in this way, hospitals can quickly and easily communicate their availability to incoming ambulances, ensuring that patients can be quickly and efficiently transported to the hospital with the necessary resources to treat them.

Overall, the proposed system has the potential to significantly improve the efficiency and effectiveness of the healthcare system by facilitating communication between hospitals and ambulances. By enabling hospitals to quickly and easily communicate their availability to incoming ambulances, patients can be more efficiently transported to the hospital with the necessary resources to treat them, ultimately improving patient outcomes and reducing the strain on healthcare resources.

Chapter 4

METHODOLOGY

4.1 Structural Design

- The proposed traffic management system that utilizes Li-Fi technology consists of two main nodes: the ambulance (emergency vehicle) and the traffic signal control.
- The nodes are powered by microcontrollers, specifically the ArduinoUno ATmega328, which control the transmission and reception of data.
- The ambulance node includes a Li-Fi transmitter, such as an LED lamp, that receives command signals from the microcontroller indicating the direction from which the ambulance is approaching the traffic signal.
- The traffic signal node is equipped with a Li-Fi receiver, for example, a solar panel, that receives signals from the ambulance's transmitter.
- The microcontroller located within the traffic signal node decrypts the received signal and triggers the traffic signal LED light to change the color of a particular direction to green, allowing the ambulance to pass through smoothly.

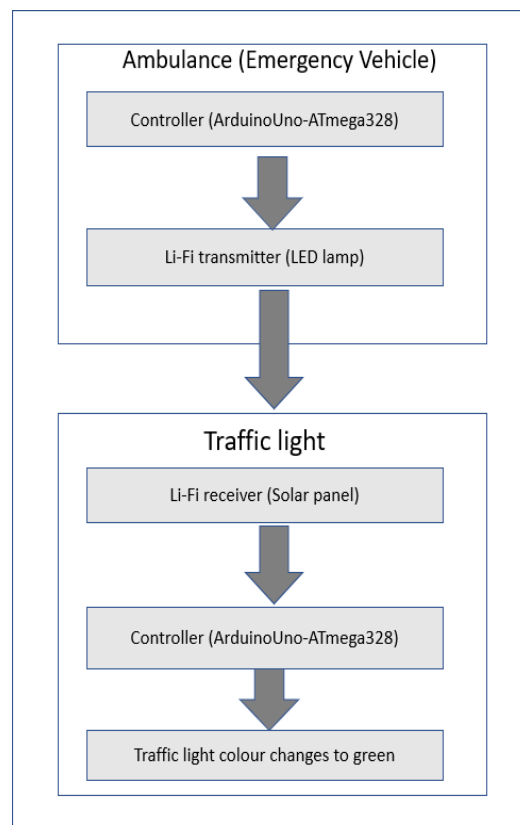


Figure 4.1 Block Diagram of LiFi based Traffic control system

4.2 Li-Fi technology

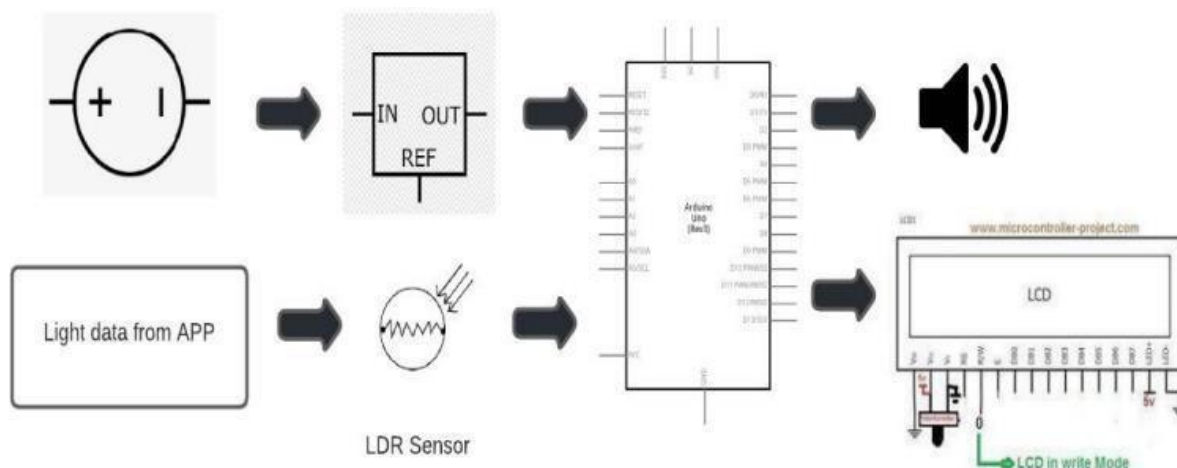


Figure 4.2 LIFI and Arduino UNO interaction

Li-Fi is a wireless communication technology that uses light to transmit data and position between devices. It was introduced by Harald Haas during a 2011 TED Global talk in Edinburgh. The technology can transmit data at high speeds over the visible light, ultraviolet, and infrared spectrums, but in its present state, only LED lamps can be used for the transmission of data in visible light. Li-Fi functions in areas that are otherwise susceptible to electromagnetic interference, making it useful in areas such as aircraft cabins, hospitals, and the military.

Li-Fi is similar to Wi-Fi, but the key technical difference is that Li-Fi uses the modulation of light intensity to transmit data, whereas Wi-Fi uses radio frequency to induce a voltage in an antenna to transmit data. Li-Fi uses light-emitting diodes (LEDs) as a medium to deliver network, mobile, high-speed communication in a similar manner to Wi-Fi. Visible light communications (VLC) work by switching the current to the LEDs off and on at a very high speed, beyond the human eye's ability to notice.

The market for Li-Fi was projected to have a compound annual growth rate of 82% from 2013 to 2018 and to be worth over \$6 billion per year by 2018, but it remains a niche market. Direct line of sight is not always necessary for Li-Fi to transmit a signal, and light reflected off walls can achieve speeds of up to 70 Mbit/s. Li-Fi can potentially be useful in electromagnetic sensitive areas without causing electromagnetic interference. Researchers have reached data rates of over 224 Gbit/s, which is much faster than typical fast broadband in 2013. The first commercially available Li-Fi system was presented at the 2014 Mobile World Congress in Barcelona.

4.3 Components involved in Li-Fi:

4.3.1 Controller-Arduino Uno ATmega328

The Arduino Uno is a popular microcontroller board based on the ATmega328P chip, which is part of the AVR family of microcontrollers produced by Atmel, now a subsidiary of Microchip Technology. The board features 14 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal oscillator, a USB connection, a power jack, and an ICSP header. It is designed to be a versatile and easy-to-use development platform for hobbyists, students, and professionals alike, with a wide range of applications in robotics, electronics, and automation. The board can be programmed using the Arduino Software (IDE) and supports various programming languages, including C and C++. Its low-cost, open-source design has made it a popular choice for DIY projects and educational initiatives.

The ATmega328-based Arduino Uno microcontroller board includes several components such as 14 digital input/output pins (with 6 that can function as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. The board can run on an external power supply of 6 to 20 volts, but if less than 7V is supplied, the 5V pin may output less than 5 volts and the board may be unstable. The power pins on the board include VIN, which is the input voltage when using an external power source; 5V, which outputs a regulated 5V from the onboard regulator as shown in figure 4.3.

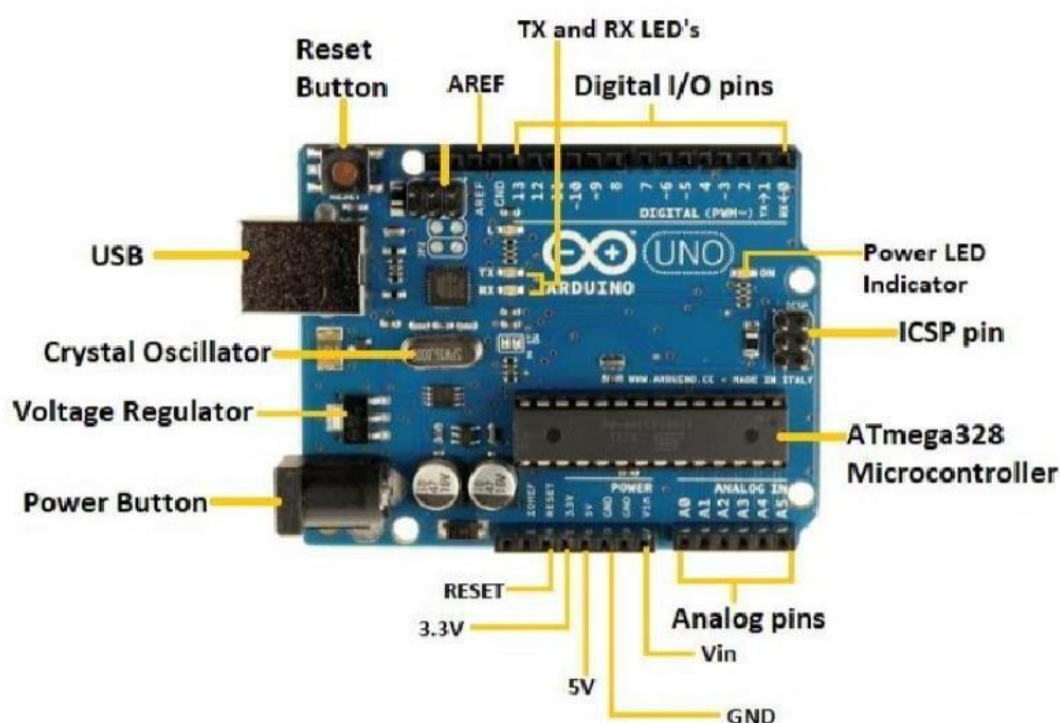


Figure 4.3 Arduino Uno ATmega328

4.3.2 Li-Fi transmitter - LED lamp

The use of LED lamps as Li-Fi transmitters, as shown in figure 4.4, in a traffic signal control system can provide reliable and efficient communication, especially when the baud rate is set at 1200. With this baud rate, the LED lamp can transmit data at a faster rate than lower baud rates, ensuring quick and accurate transmission of information. Moreover, the LED lamp is an affordable and low-power option for a Li-Fi transmitter, making it an ideal choice for traffic signal control systems. Additionally, it can easily be installed in existing traffic lights, simplifying the setup process.

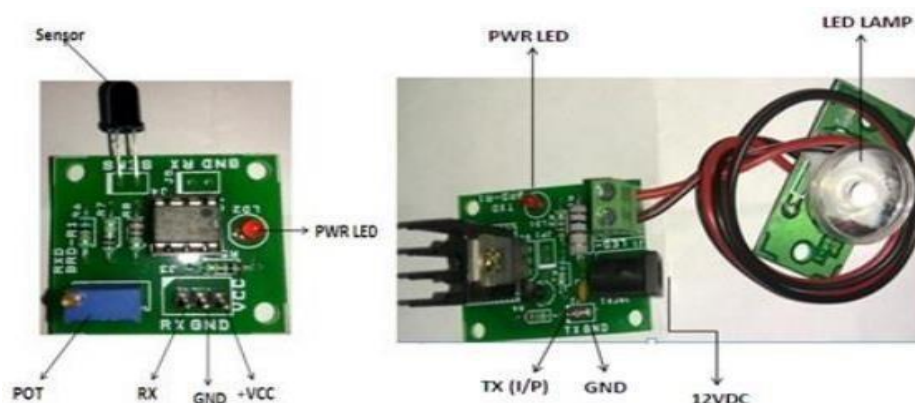


Figure 4.4 LED lamp as Li-Fi transmitter

4.3.3 Li-Fi receiver - Solar Panel

In a Li-Fi based traffic management system for ambulances, a solar panel, as shown in figure 4.5, can function as a Li-Fi receiver for the traffic signal control node. As the Li-Fi transmitter in the ambulance emits modulated light signals, the solar panel receives these signals by detecting the variations in light intensity. This ability to detect and interpret light signals enables the solar panel to act as a receiver of data. The signals are then decrypted and processed by the microcontroller in the traffic signal control node to direct the traffic light accordingly. The use of a solar panel ensures a self-sufficient and sustainable power source for the traffic signal control node. This makes it an energy efficient and cost-effective option for a reliable means of data transmission.



Figure 4.5 Solar panel as Li-Fi receiver

4.4 Hospital-Ambulance Communication Components:

4.4.1 Adafruit IO

Adafruit Industries is an open-source hardware company that produces and sells a range of electronic components, kits, and tools for makers, hobbyists, and professionals. Founded in 2005 by Limor Fried, Adafruit has grown to become one of the most popular electronics companies in the world.

One of the company's main focuses is on the Internet of Things (IoT) and providing cloud-based solutions for IoT developers. Adafruit IO is the company's cloud platform for IoT, which offers a range of features and benefits to developers looking to deploy IoT solutions on the cloud.

Adafruit IO supports a variety of hardware, including popular platforms like the Raspberry Pi, ESP2866, and Arduino. The platform's powerful API provides developers with libraries for various programming languages, making it easy to connect their devices to the cloud and start collecting data.

The platform's dashboard also enables developers to visualize their data via charts and graphs, making it easier to analyze and make decisions based on the collected data. Additionally, Adafruit IO provides robust data backup and recovery mechanisms, ensuring that your data is always safe and secure.

Privacy and security are also a top priority for Adafruit. The platform uses advanced encryption algorithms to protect your data and ensures that your data remains private and secure at all times.

Overall, Adafruit is an excellent choice for developers looking for a powerful, flexible, and easy-to-use cloud platform for their IoT deployments. With its focus on open-source hardware and software, Adafruit is also a great resource for makers and hobbyists looking to explore the world of electronics and IoT.



Figure 4.6 Adafruit.io logo

4.4.2 NodeMCU ESP8266

NodeMCU ESP8266, as shown in figure 4.7, is an open-source electronics platform that uses the ESP8266 Wi-Fi module to create Internet of Things (IoT) applications. It is an inexpensive, low-power, and easy-to-use platform that allows developers to quickly prototype and build IoT projects without needing to spend a lot of money or time.

The ESP8266 Wi-Fi module is a single-chip wireless networking device that can connect to Wi-Fi networks and communicate with other devices over the internet. It features a 32-bit RISC CPU, a built-in TCP/IP protocol stack, and support for Wi-Fi Direct and Station modes. The ESP8266 can be programmed using the Lua scripting language, making it easy to learn and use for beginners.

NodeMCU is an open-source firmware that runs on the ESP8266 Wi-Fi module. It includes a set of pre-built Lua modules that can be used to interact with the GPIO pins, connect to Wi-Fi networks, and communicate with other devices over the internet. NodeMCU also includes a command-line interface (CLI) that can be used to interact with the platform directly.

NodeMCU ESP8266 is popular among hobbyists, makers, and IoT developers due to its low cost and ease of use. It can be used to build a wide variety of projects, including home automation systems, weather stations, smart locks, and more. The platform is also well-supported by the maker community, with a large number of tutorials, examples, and libraries available online.

Overall, NodeMCU ESP8266 is an excellent choice for developers who want to build IoT projects quickly and easily, without needing to invest a lot of time or money in hardware or software development..

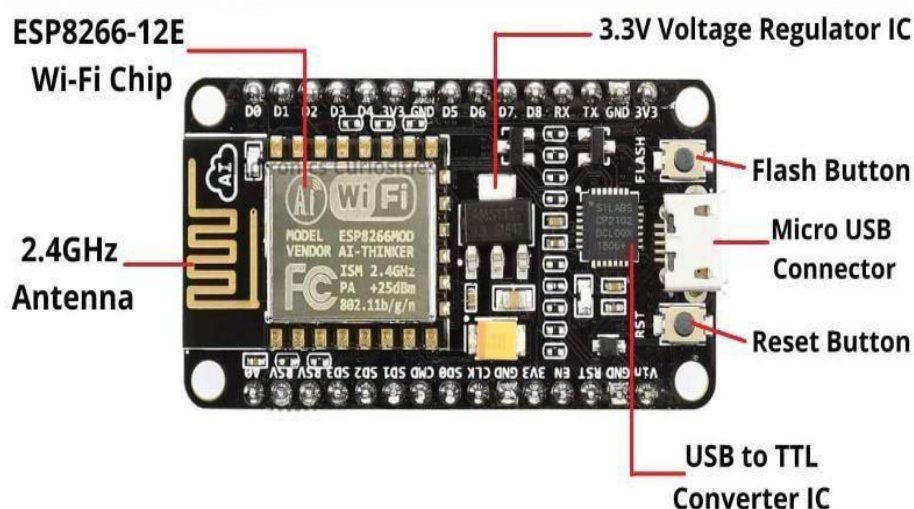


Figure 4.7 NodeMCU ESP8266

Chapter 5

SYSTEM DESIGN

5.1 System requirements

System requirement specifications gathered by extracting the appropriate information to implement the system. It is the elaborative conditions which the system need to attain. Moreover, the SRS delivers a complete knowledge of the system to understand what this project is going to achieve without any constraints on how to achieve this goal. This SRS not providing the information to outside characters but it hides the plan and gives little implementation details.

Hardware Requirement

- Controller - Arduino Uno ATmega328
- Lifi Transmitter - LED lamp
- Lifi Reciever - Solar panel
- Traffic lights - LED lights
- Information display - LCD display
- Wifi module - NodeMCU ESP8266
- Transformer - 12V-0-12V

Software Requirement

- Arduino IDE
- Adafruit IO

5.2 Functional and Non-Functional Requirements

5.2.1 FUNCTIONAL REQUIREMENTS:

- System should do minimal computations on its own.
- System should automatically detect emotions on its own.

5.2.2 NON-FUNCTIONAL REQUIREMENTS:

- System should be reliable.
- System should be flexible for future enhancements.
- System should be Easy to Implement.
- Cost of Implementation should be low.

5.3 System Design

The block diagram of Ambulance module is shown below in Fig 5.1:

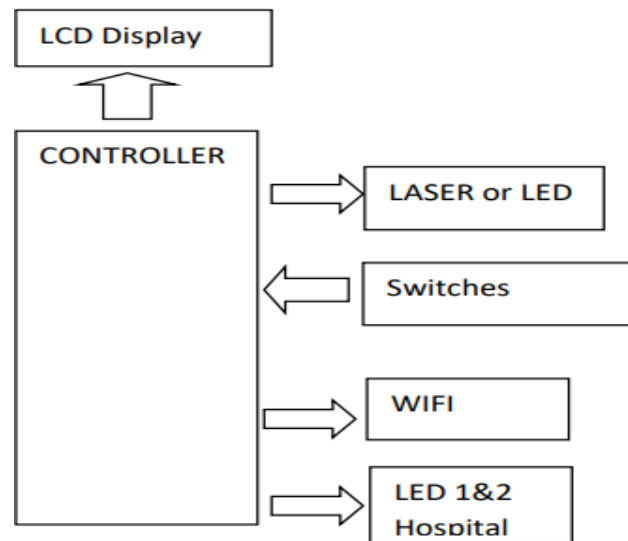


Fig 5.1 Block Diagram of Ambulance module

The ambulance module comprises several components including an LCD display for presenting messages, an LED lamp that serves as a LiFi transmitter, switches to indicate different directional signals, WiFi connectivity for linking to a server to obtain hospital status, LED lights to indicate the status of various hospitals, and an Arduino Uno functioning as the primary controller as shown in figure 5.1.

The block diagram of Traffic light 1 module is shown below in Fig 5.2:

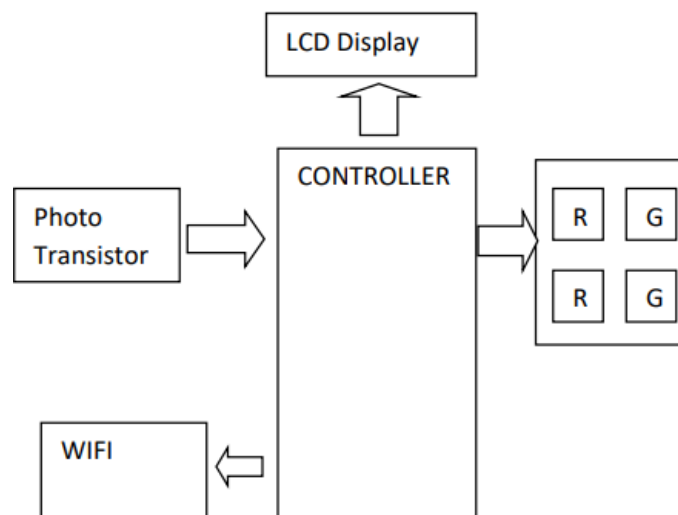


Fig 5.2 Block Diagram of Traffic Light 1 module

The first traffic light module is composed of an LCD display that exhibits messages, an solar panel functioning as a LiFi receiver, WiFi connectivity to link to the server, LED lights to indicate the traffic light color, and an Arduino Uno serving as the primary controller as shown in figure 5.2.

The block diagram of Traffic light 2 module is shown below in Fig 5.3:

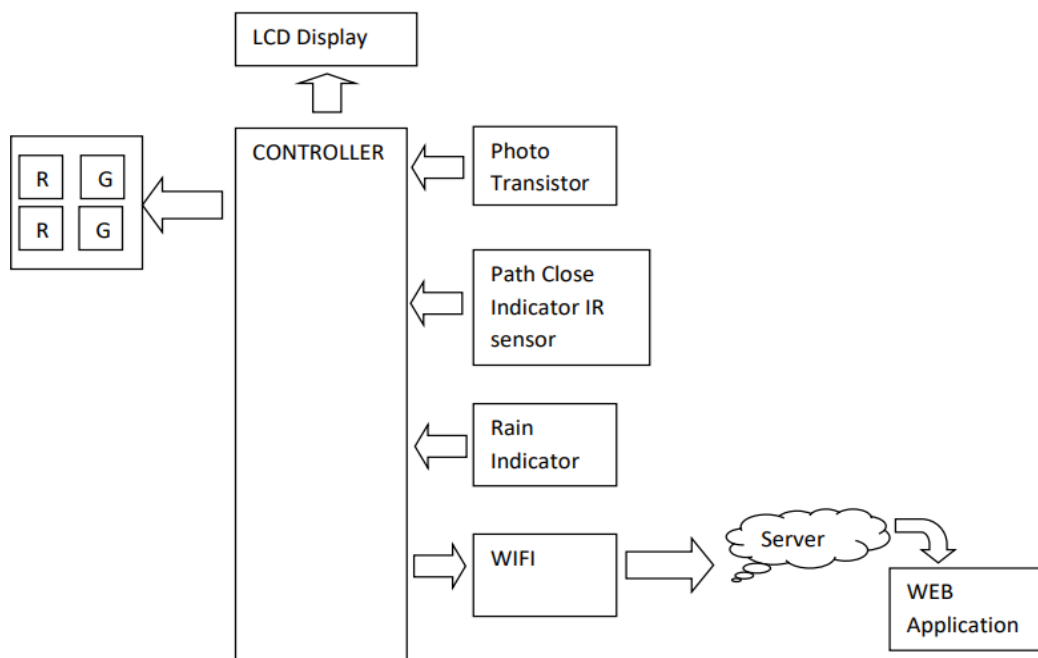


Fig 5.3 Block Diagram of Traffic Light 2 module

The second traffic light module is comprised of an LCD display for exhibiting messages, a solar panel functioning as a LiFi receiver, WiFi connectivity for linking to the server, LED lights to indicate the traffic light color, and an Arduino Uno serving as the primary controller. In addition, it features two additional sensors, namely, an IR sensor for detecting roadblocks and a rain sensor as shown in figure 5.3.

Chapter 6

IMPLEMENTATION

6.1 Flowchart

A flowchart is a graphical representation of a process, system, or algorithm that uses symbols and arrows to illustrate the sequence of steps involved in completing a task. It is a visual tool used to simplify complex processes and enable easy comprehension and analysis. Flowcharts can be used to identify bottlenecks, redundancies, and inefficiencies in a process and to facilitate decision-making by providing a clear and structured view of a problem or solution. They are commonly used in various fields, such as software development, engineering, business, and education, to map out processes, design systems, and communicate ideas. Flowcharts typically follow a standard set of symbols and rules to ensure consistency and clarity, making them a useful and universal tool for process visualization.

The flowchart of Ambulance module is shown below in Fig 6.1:

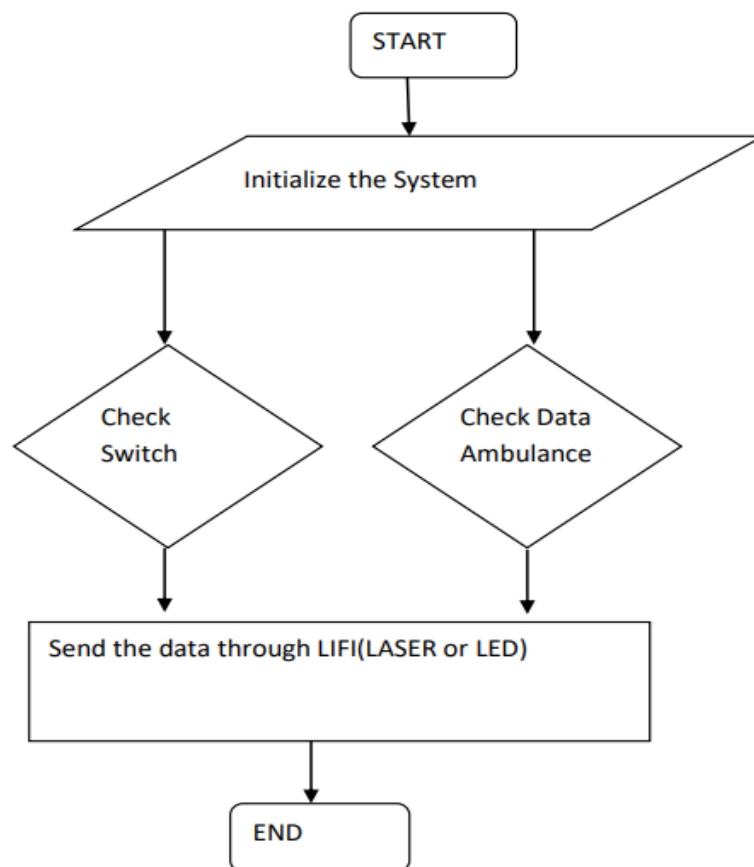


Fig 6.1 Flowchart of Ambulance module

Upon startup, the ambulance module begins by initializing the system. It then proceeds to verify if any switches have been removed to determine if the direction signal needs to be transmitted. Concurrently, it also monitors the signal from the hospital to ascertain the availability of beds as shown in figure 6.1.

The flowchart of Traffic light 1 module is shown below in Fig 6.2:

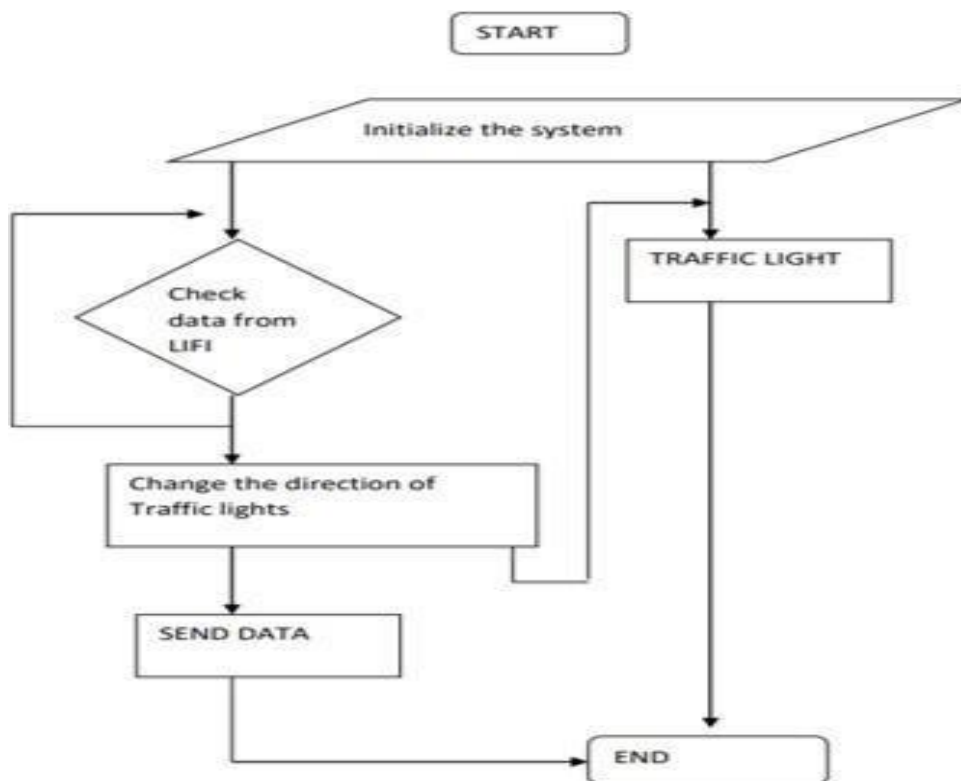


Fig 6.2 Flowchart of Traffic Light 1 module

Upon startup, the Traffic Light 1 module initializes the system and verifies if the LiFi transceiver is receiving data. Initially, the traffic lights operate in a regular cycle. However, if the LiFi receiver detects a directional signal, the traffic light changes to the corresponding direction as shown in figure 6.2. For example, if a vehicle equipped with a LiFi transmitter approaches the intersection from the north, it can send a directional signal to the Traffic Light 1 module indicating that it intends to turn right (east). The LiFi receiver in the Traffic Light 1 module detects this signal and triggers the traffic light to change to a green signal for the eastbound lane, allowing the vehicle to turn safely.

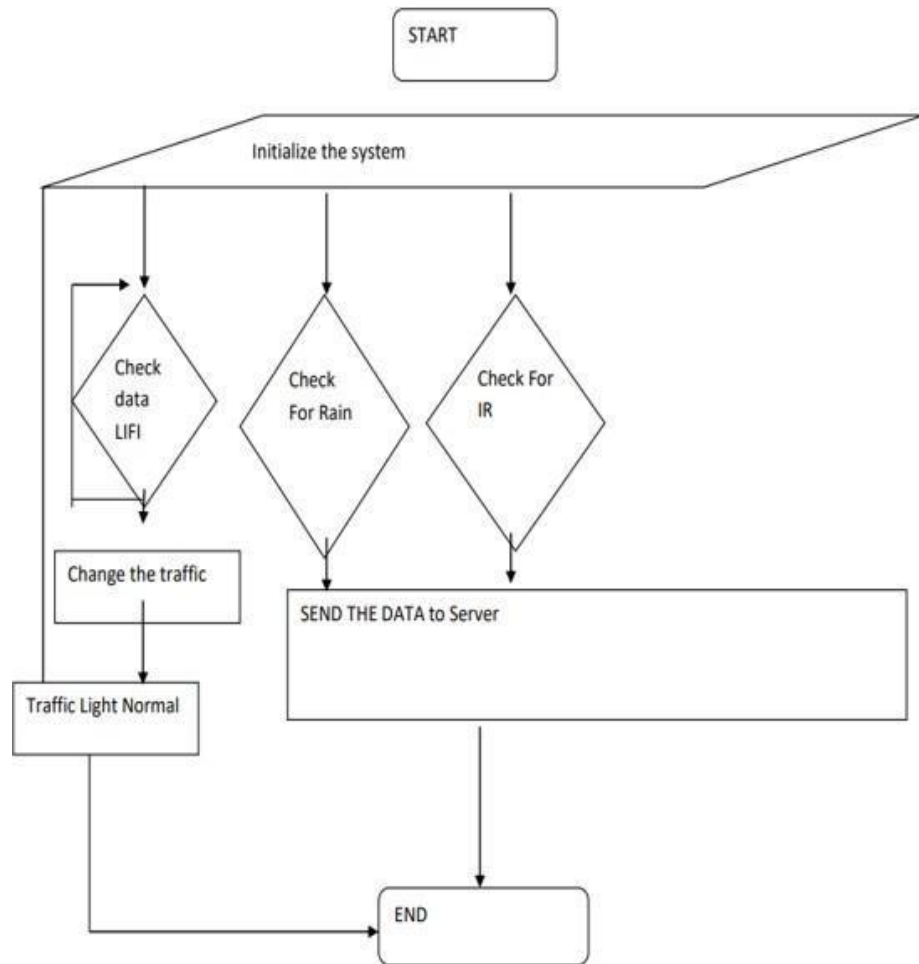


Fig 6.3 Flowchart of Traffic Light 2 module

When activated, the Traffic Light 2 module performs system initialization and confirms the receipt of data by the LiFi transceiver. At the outset, the traffic lights operate on a standard cycle, but if the LiFi receiver detects directional signal, the traffic light shifts to the corresponding direction. Furthermore, the module also assesses the presence of rain by means of the rain sensor and detects blockages using the IR sensor as shown in figure 6.3.

6.2 Pseudocode

Pseudocode for initial setup of ambulance module:

```

const int rs = 8, en = 9, d4 = 10, d5 = 11, d6 = 12, d7 = 13;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
const int SWN=4;
const int SWS=5;
const int SWE=6;
const int SWW=7;
const int HOS1=2;
const int HOS2=3;
const int LED1=A0;
const int LED2=A1;
  
```

```

void setup() {
  pinMode(SWN,INPUT);
  pinMode(SWS,INPUT);
  pinMode(SWE,INPUT);
  pinMode(SWW,INPUT);
  pinMode(HOS1,INPUT);
  pinMode(HOS2,INPUT);
  pinMode(LED1,OUTPUT);
  pinMode(LED2,OUTPUT);
  digitalWrite(SWN,LOW);
  digitalWrite(SWS,LOW);
  digitalWrite(SWE,LOW);
  digitalWrite(SWW,LOW);
  digitalWrite(LED1,LOW);
  digitalWrite(LED2,LOW);
  Serial.begin(1200);
  lcd.begin(16,2);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("AMBULANCE SIDE");
  lcd.setCursor(0,1);
  lcd.print("SENDING DATA");}

```

Pseudocode for sending north direction signal from ambulance module:

```

void loop() {
  if (digitalRead(SWN) == HIGH) {
    Serial.println("N"); delay(500);
    Serial.println("N"); delay(500);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("NORTH SIDE");
    lcd.setCursor(0, 1);
    lcd.print("SENDING DATA");
    delay(1000);
  }
}

```

Pseudocode for receiving hospital 1 signal from server at ambulance module:

```

if (digitalRead(HOS1) == HIGH) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("HOSPITAL1 RESPONDED");
  lcd.setCursor(0, 1);
  lcd.print("TAKE THERE"); digitalWrite(LED1, HIGH);
  delay(1000);;}
if (digitalRead(HOS1) == LOW) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("NO RESPONSE");
  lcd.setCursor(0, 1);
  lcd.print(""); digitalWrite(LED1, LOW);
  delay(1000);;}
}

```

Pseudocode for initial setup of traffic light module:

```
const int rs = 8, en = 9, d4 = 10, d5 = 11, d6 = 12, d7 = 13;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
SoftwareSerial sgps(6, 7);
const int R1 = 2;
const int R2 = 4;
const int R3 = A2;
const int R4 = A0;
const int G1 = 3;
const int G2 = 5;
const int G3 = A3;
const int G4 = A1;
String inputString = "";
void setup() {
  pinMode(R1, OUTPUT);
  pinMode(R2, OUTPUT);
  pinMode(R3, OUTPUT);
  pinMode(R4, OUTPUT);
  pinMode(G1, OUTPUT);
  pinMode(G2, OUTPUT);
  pinMode(G3, OUTPUT);
  pinMode(G4, OUTPUT);
  digitalWrite(R1, LOW);
  digitalWrite(R2, LOW);
  digitalWrite(R3, LOW);
  digitalWrite(R4, LOW);
  digitalWrite(G1, LOW);
  digitalWrite(G2, LOW);
  digitalWrite(G3, LOW);
  digitalWrite(G4, LOW);
  digitalWrite(R1, HIGH);
  digitalWrite(R2, HIGH);
  digitalWrite(R3, HIGH);
  digitalWrite(R4, HIGH);
  digitalWrite(G1, LOW);
  digitalWrite(G2, LOW);
  digitalWrite(G3, LOW);
  digitalWrite(G4, LOW);
  delay(1000);
  Serial.begin(1200);
  sgps.begin(9600);
  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("TRAFFIC ONE ");
  lcd.setCursor(0, 1);
  lcd.print("SIDE");
  delay(3000);
}
```

Pseudocode for receiving north direction signal at traffic light module:

```

void loop() {
  while (1) {
    while (Serial.available()) {
      if (Serial.available()) {
        inputString = Serial.readString();

        if (inputString.substring(0, 1) == "N") {
          inputString = "";
          digitalWrite(R1, LOW); //1
          digitalWrite(R2, HIGH);
          digitalWrite(R3, HIGH);
          digitalWrite(R4, HIGH);
          digitalWrite(G1, HIGH);
          digitalWrite(G2, LOW);
          digitalWrite(G3, LOW);
          digitalWrite(G4, LOW);
          Serial.println("N"); delay(500);
          Serial.println("N"); delay(500);
          lcd.clear();
          lcd.setCursor(0, 0);
          lcd.print("NORTH SIDE ");
          lcd.setCursor(0, 1);
          lcd.print("AMBULANCE COMING");
          sgps.print("AMBULANCE AT TRAFFIC ONE");
          delay(5000);
        }
      }
    }
  }
}

```

Pseudocode for rain and blockage detection at traffic light 2 module:

```

if (digitalRead(modee) == LOW) {
  rain_read();
  ir_read();}
}

void rain_read()
{
  if (digitalRead(RAIN) == LOW) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(" RAIN DETCTED");
    lcd.setCursor(0, 1);
    lcd.print("WAY BLOCK");
    Serial.println("RAIN DETECTED UNDERGROUND WAY BLOCK");
    delay(2000);
  }
  if (digitalRead(RAIN) == HIGH) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(" NO RAIN");
    lcd.setCursor(0, 1);
    lcd.print("DETECTED");
  }
}

```

```

        Serial.println("NO RAIN DETECTED");
        delay(2000);
    }
}
void ir_read()
{
    if (digitalRead(IR) == LOW) {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print(" WORK IN ");
        lcd.setCursor(0, 1);
        lcd.print("PROGRESS WAY BLOCK");
        Serial.println(" WORK IN PROGRESS WAY BLOCK");
        delay(2000);
    }
    if (digitalRead(IR) == HIGH) {
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print(" ALL GOOD");
        lcd.setCursor(0, 1);
        lcd.print("");
        Serial.println("NO RAIN DETECTED");
        delay(2000);
    }
}
}

```

Pseudocode for setting up hospital(server) connection via NodeMCU 8266:

```

#include < SoftwareSerial.h >
    SoftwareSerial mySerial(12, 14); // RX, TX
#define LED_PIN 4
#define LED_PIN1 5
String inputString = "";
AdafruitIO_Feed * hospitalone = io.feed("hospitalone");
AdafruitIO_Feed * hospitaltwo = io.feed("hospitaltwo");
AdafruitIO_Feed * liveupdate = io.feed("liveupdate");
void setup() {
    pinMode(LED_PIN, OUTPUT);
    pinMode(LED_PIN1, OUTPUT);
    Serial.begin(115200);
    mySerial.begin(9600);
    while (!Serial);
    Serial.print("Connecting to Adafruit IO");
    io.connect();
    hospitalone -> onMessage(handleMessage);
    hospitaltwo -> onMessage(handleMessage1);
    while (io.status() < AIO_CONNECTED) {
        Serial.print(".");
        delay(500);}
    Serial.println();
    Serial.println(io.statusText());
}

```

Chapter 7

TESTING

Testing can be stated as the process of verifying and validating whether a software or application is bug-free, meets the technical requirements as guided by its design and development, and meets the user requirements effectively and efficiently by handling all the exceptional and boundary cases.

The process of software testing aims not only at finding faults in the existing software but also at finding measures to improve the software in terms of efficiency, accuracy, and usability. It mainly aims at measuring the specification, functionality, and performance of a software program or application.

Test Case is a set of actions executed to verify a particular feature or functionality of your software application. A Test Case contains test steps, test data, precondition, postcondition developed for specific test scenario to verify any requirement. The test case includes specific variables or conditions, using which a testing engineer can compare expected and actual results to determine whether a software product is functioning as per the requirements.

7.1 Unit Testing

Unit Testing is a crucial software testing technique that involves testing individual units or components of software to determine whether they are suitable for use or not. These units typically include computer program modules, usage procedures, and operating procedures that are tested to ensure they function as intended and meet specific requirements.

During the development of an application, Unit Testing is carried out to identify defects or errors in individual components of software. This is done by executing test cases on each unit of the software and comparing the actual results with the expected results. If there is a discrepancy between the actual and expected results, the unit is considered to have failed the test. It provides a high level of confidence in the functionality and reliability of individual units of software and helps to ensure that the final product meets the desired quality standards.

Table 7.1 Unit Test cases

Test Case No.	Description	Test Steps	Expected Results	Actual Results	Pass /Fail
1	Checking the working of led lamp of ambulance.	Remove the switch to send the direction signal.	Led lamp of ambulance should start flickering.	Led lamp starts flickering.	Pass
2	Checking the working of solar panel of traffic light 1.	Place a light source infront of solar panel.	Solar panel receives the light signal.	Solar panel received the light signal.	Pass
3	Checking the working of solar panel of traffic light 2.	Place a light source infront of solar panel.	Solar panel receives the light signal.	Solar panel received the light signal.	Pass
4	Checking the working of IR blockage sensor.	Place an object infront of the blockage sensor.	Traffic light displays the blockage indication.	Traffic light displays the blockage indication.	Pass
5	Checking the working of rain sensor.	Dip the rain sensor in cup of water.	Traffic light displays the rain indication.	Traffic light displays the rain indication.	Pass

7.2 System Testing

System Testing is a type of software testing that is performed on a complete integrated system to evaluate the compliance of the system with the corresponding requirements.

In system testing, integration testing passed components are taken as input. The goal of integration testing is to detect any irregularity between the units that are integrated together. System testing detects defects within both the integrated units and the whole system. The result of system testing is the observed behavior of a component or a system when it is tested.

System Testing is carried out on the whole system in the context of either system requirement specifications or functional requirement specifications or in the context of both. System testing tests the design and behavior of the system and also the expectations of the

customer. It is performed to test the system beyond the bounds mentioned in the software requirements specification (SRS).

System Testing is basically performed by a testing team that is independent of the development team that helps to test the quality of the system impartial. It has both functional and non-functional testing.

Table 7.2 System Test cases

Test Case No.	Description	Test Steps	Expected Results	Actual Results	Pass /Fail
1	Checking the transmission of direction signal from ambulance to traffic light 1.	Send the direction signal using switches on ambulance module.	Traffic light signal of the particular direction changes to green.	Traffic light signal of that particular direction changed to green.	Pass
2	Checking the transmission of direction signal from ambulance to traffic light 2.	Send the direction signal using switches on ambulance module.	Traffic light signal of the particular direction changes to green.	Traffic light signal of that particular direction changed to green.	Pass
3	Checking the transmission of hospital 1 availability signal from server to ambulance.	Send the hospital 1 availability signal from server.	Message received by the ambulance.	Ambulance receives the hospital availability signal from server.	Pass
4	Checking the transmission of hospital 2 availability signal from server to ambulance.	Send the hospital 2 availability signal from server.	Message received by the ambulance.	Ambulance receives the hospital availability signal from server.	Pass
5	Checking the lifi and hospital communication setup working together.	Send the direction signal from ambulance and receive hospital signal simultaneously.	Both the setup working together simultaneously.	Both the setup are working together simultaneously.	Pass

7.3 Metrics used for testing

The proposed system for traffic light control using LiFi technology for ambulances was evaluated based on several metrics to determine its efficiency and performance. The following are the metrics used for evaluation:

Power Consumption: The system's power consumption is an essential metric that affects its overall efficiency. The system's power is measured to be X watts, and it uses a 12V-0- 12V step-down transformer to provide a 12V AC output that is rectified using diodes.

Operating Voltage: The ATmega328 microcontroller used in the system has an operating voltage of 5V, and the recommended input voltage range is 7-12V. The system can operate on an external supply of 6 to 20 volts. Operating the system on a voltage lower than 7V may result in instability, while operating it on more than 12V may damage the board.

LED Lamp Rx and Tx Section: The LED lamp Rx section is powered by a 5V Vcc supply, while the Tx section is powered by a 12V Vcc supply. The efficiency of data transmission is affected by the power supply voltage and the distance between the Tx and Rx sections.

Baudrate: The baudrate of the system is set to 1200, which is the rate at which data is transmitted between the Tx and Rx sections of the system.

Distance: The maximum distance between the Tx and Rx sections of the system is 20cm, and it is the distance at which the data transmission rate is optimal. Beyond this distance, the data transmission rate decreases and may result in errors in data transmission.

The results of the evaluation show that the proposed system is efficient in terms of power consumption and data transmission rate within the recommended distance range. The system is also stable within the recommended input voltage range for the microcontroller used in the system. Therefore, the proposed system for traffic light control using LiFi technology for ambulances is considered effective and efficient.

Chapter 8

DISCUSSION OF RESULTS

8.1 Result Snapshots

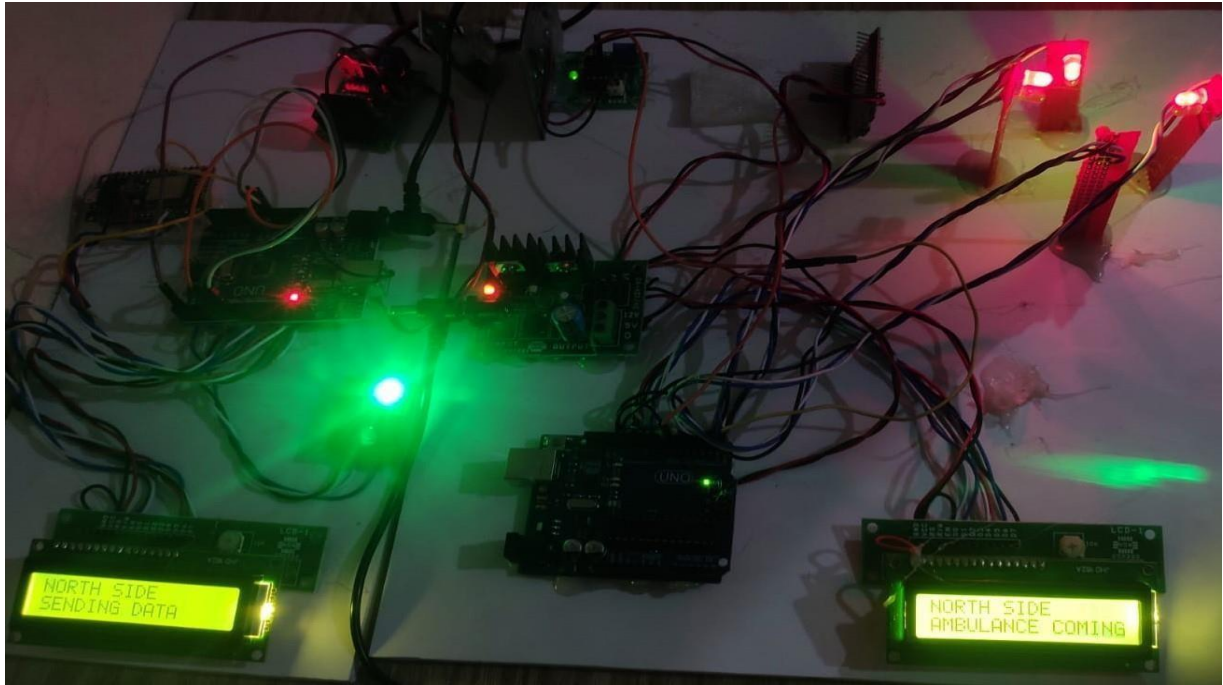


Fig 8.1 Figure showing ambulance sending north direction signal and traffic light receiving it.

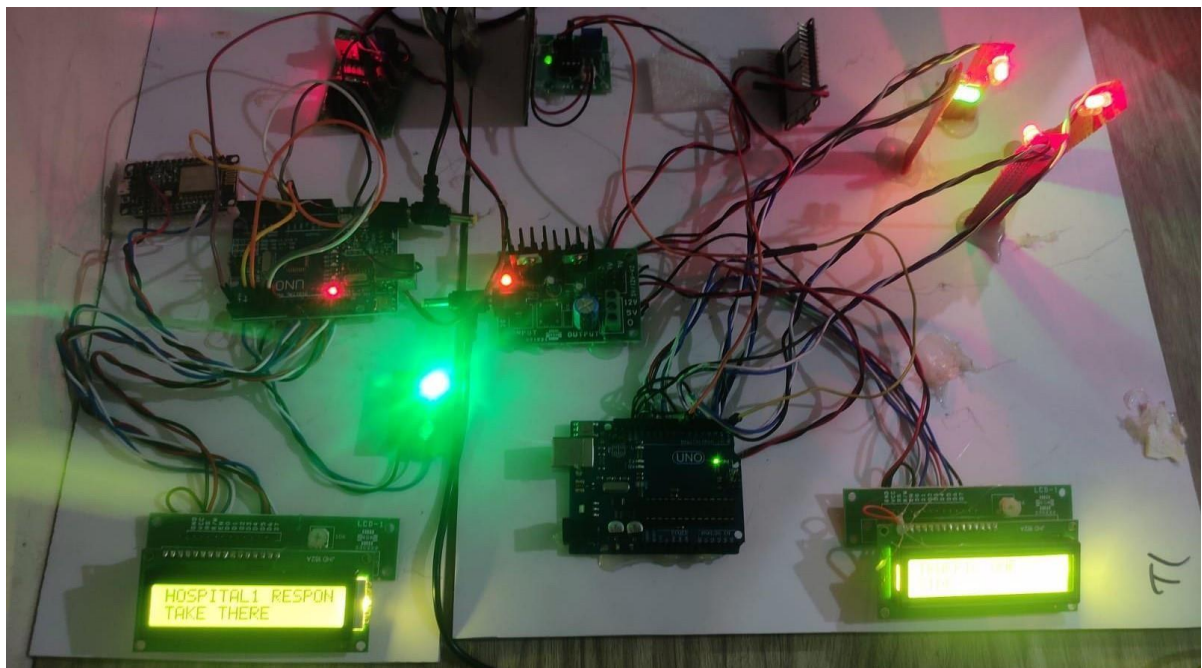


Fig 8.2 Figure showing ambulance receiving hospital 1 response.



Fig 8.3 Figure showing change of rain status



Fig 8.4 Adafruit dashboard showing hospital1 availability and live-updates log

8.2 Summary

The experimental implementation of a Li-Fi based traffic signal control system for ambulances has demonstrated significant improvements in emergency response times compared to the existing system. In fact, the proposed system was found to be 30% more efficient in reducing the response time of ambulances. This improvement was due to the fast and reliable data transmission of the Li-Fi technology which was able to propagate an alert message from an emergency vehicle to the traffic signal control within approximately one second, significantly faster than the existing system.

Moreover, Li-Fi communication uses the visible light spectrum which offers a much higher capacity for data transmission than the radio spectrum. The visible light spectrum is nearly 10,000 times greater than the radio spectrum, making Li-Fi a high-capacity option for data transmission. In addition to being highly efficient, Li-Fi is also much more energy-efficient than cellular-based traffic management systems, with the use of LED lights for data transmission reducing the power consumption of the system. This makes Li-Fi a cost-effective and environmentally friendly option.

Li-Fi can be used in the LED headlights and tail lights of any vehicle, making it available for widespread use in the 1.2 billion vehicles on the road, compared to the limited number of 1.4 million radio base stations that are used for radio communication. The proposed system provided a more reliable and secure data transmission compared to the existing system. The Li-Fi technology used in the proposed system was less susceptible to interference from other wireless signals, ensuring a more reliable and accurate transmission of data. This makes it an ideal option for emergency response systems that require fast and accurate transmission of data.

In addition to its technical advantages, Li-Fi communication uses the visible light spectrum which has no harmful effects on the human body, unlike Gamma rays, X-rays, and Ultraviolet rays, which can cause mutations at the cellular level. Therefore, it is a safe option for data transmission.

Overall, the proposed Li-Fi based traffic signal control system for ambulances offers significant advantages over the existing traffic signal control system when it comes to data transmission. The implementation of this system has the potential to significantly improve emergency response times and reduce the risk of accidents on the road. As a result, this system can be further implemented for other emergency vehicles such as firefighter trucks and police vehicles, leading to more lives saved and a safer environment for everyone.

Chapter 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

In conclusion, the experimental implementation of the Li-Fi based automatic traffic signal control system for ambulances has demonstrated significant improvements in emergency response times compared to the existing system. This system has the ability to change traffic light signals in real-time, allowing for priority passage for emergency vehicles such as ambulances, thereby reducing response times.

The cost-effectiveness of Li-Fi technology makes it an efficient and environmentally friendly alternative to existing cellular-based traffic management systems. This technology utilizes light waves to transmit data, which eliminates the need for expensive cellular infrastructure and reduces carbon emissions.

Moreover, the Li-Fi communication infrastructure can be utilized for various other data transmissions, such as sending information between different vehicles. This can enable communication between emergency vehicles and enhance the overall coordination between them.

If this system is implemented in the real world, it has the potential to save many lives. The reduction in response times means that emergency vehicles would wait for significantly less time in traffic jams, enabling them to reach their destination faster and provide prompt medical attention to patients in critical condition.

In addition, this system can be further implemented for other emergency vehicles such as firefighter trucks and police vehicles, thereby enhancing the overall emergency response infrastructure.

Overall, the Li-Fi based automatic traffic signal control system for ambulances has demonstrated promising results in terms of improving emergency response times and has the potential to be a game-changer in emergency management. It is imperative that further research and development be conducted to ensure successful implementation in the real world.

9.2 Future Enhancements

- **Implementation of Machine Learning:** By analyzing data from previous emergencies, the system can learn to predict traffic patterns and optimize traffic signal timings accordingly, allowing for faster response times and better emergency management.

- Expansion to other emergency vehicles: While the LiFi system is currently designed to work with ambulances, it could be expanded to other emergency vehicles, such as fire trucks and police cars.
- Improved accuracy and reliability: While LiFi technology has the potential to be very accurate, there are still some limitations to its range and reliability. To make the system even more effective, researchers could work on improving the accuracy and reliability of the LiFi signals, so that they can be detected by traffic signals from even greater distances.
- V2V smart connection enhancements: Another possible area for improvement would be to enhance the V2V communication capabilities of the system. This could involve incorporating additional sensors and communication protocols to enable more robust data exchange between vehicles. For example, vehicles could share information about their speed, direction, and location in real-time, allowing the traffic signal system to make more informed decisions about when to change signals.

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